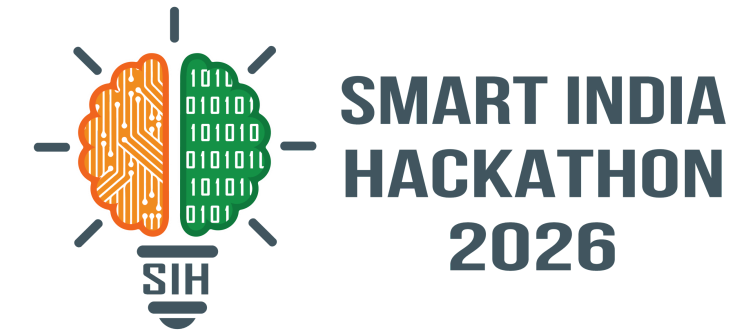


SMART INDIA HACKATHON 2026

JEEVAN SETU



Problem Statement ID – SIH26039

**Problem Statement Title – AI-Powered Underground Mine Safety,
Monitoring and Rescue System**

Theme – Smart Automation

PS Category – Hardware

Team Name – MINERDS



Smart Helmet

Provides real-time data and communication for rescuers.

Rescue Rover

Navigates hazardous terrain and delivers supplies.

Control Station

Manages operations and coordinates rescue efforts.

“When danger strikes underground, information saves lives.”

UNDERSTANDING THE PROBLEM

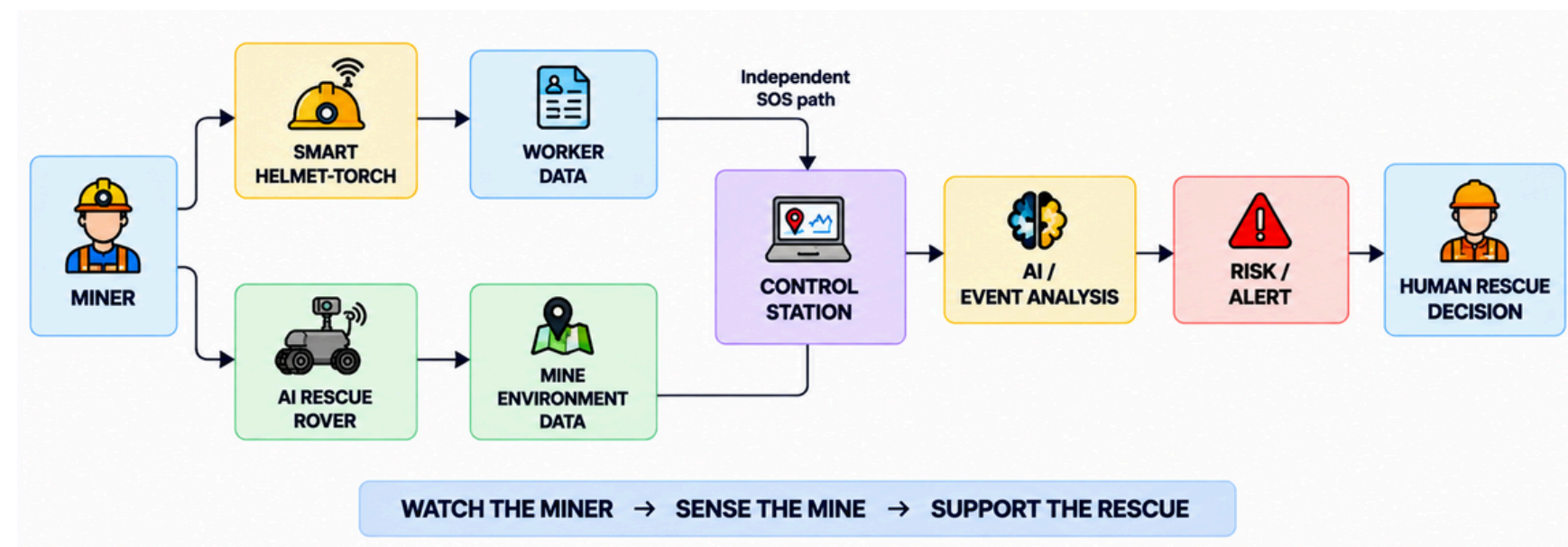
- **LIMITED SITUATIONAL AWARENESS:** Rescue teams may lack a live picture of mine conditions and worker status.
- **WORKER LOCATION:** Poor visibility and disrupted communication make underground localization difficult.
- **HIDDEN HAZARDS:** Gas, heat, flooding and debris can make direct reconnaissance dangerous.
- **DELAYED EMERGENCY RESPONSE:** Critical worker and environmental information may not reach the control room quickly enough.
- **ROOT CAUSE:** Fragmented underground information → delayed and uncertain rescue decisions

OUR SOLUTION

- **Smart Helmet & Torch Module:** Enhances the miner's existing safety helmet/torch with worker ID, location, movement/fall detection, vital monitoring and an emergency SOS button.
- **AI Rescue Rover:** Enters hazardous or inaccessible areas to collect gas, environmental, visual, thermal and LiDAR data without exposing rescuers to unnecessary danger.
- **Unified Control Station:** Combines worker + rover data on a single laptop dashboard for real-time monitoring and emergency coordination.
- **Intelligent Alerts:** Correlates worker distress, location and mine conditions to identify high-risk situations and generate priority alerts.
- **Human-in-the-Loop Rescue:** Provides rescue teams with real-time situational information and recommendations, while the final rescue decision remains with the human operator.

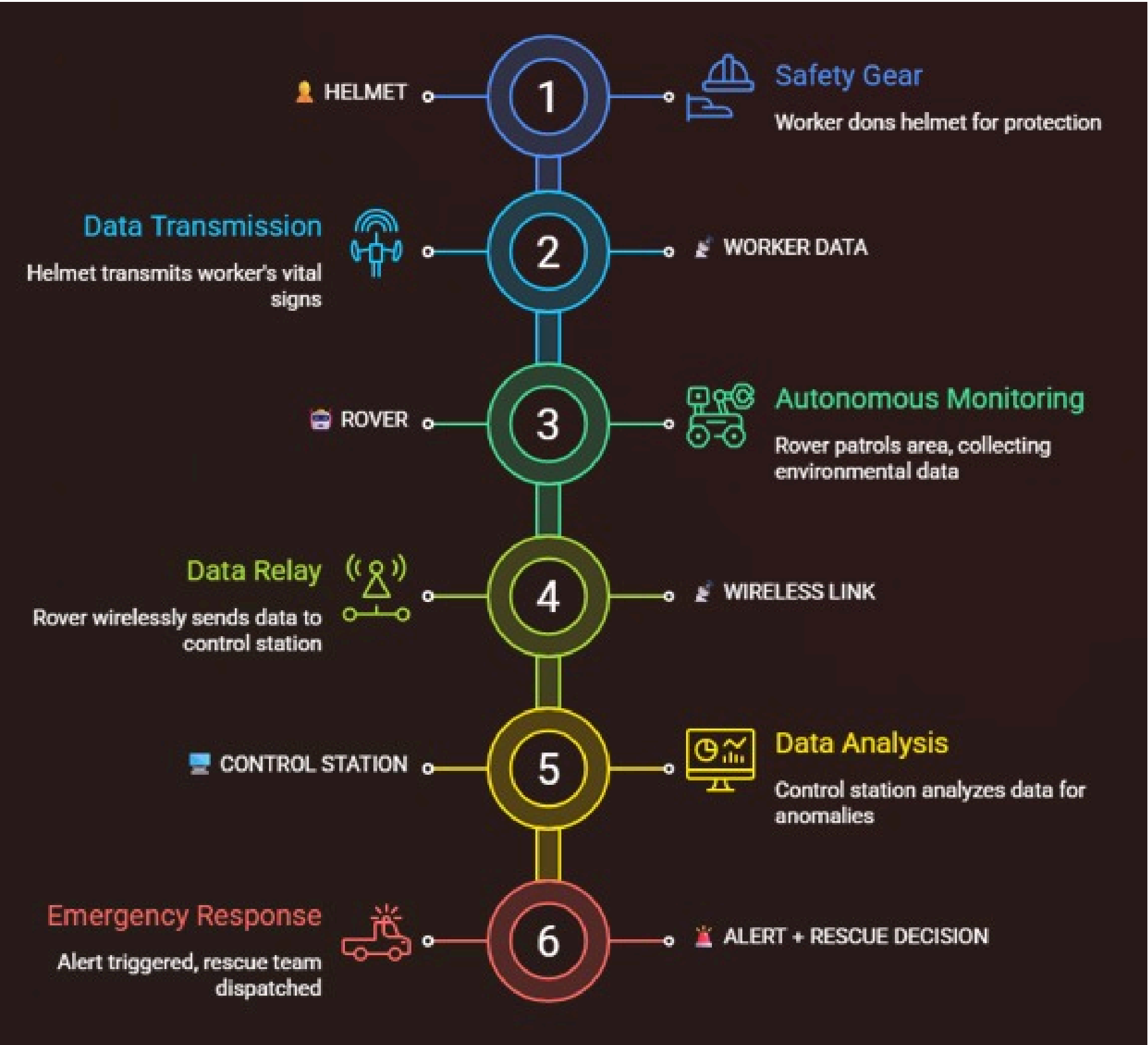
OUR UNIQUE SOLUTIONS

- **EXISTING-EQUIPMENT INTEGRATION:** Adds intelligence to the miner's existing helmet/torch instead of another wearable.
- **DIRECT WORKER SOS:** Immediate emergency signal from the worker to the control room.
- **MOBILE MINE INTELLIGENCE:** Rover remotely captures environmental and visual information from hazardous zones.
- **WORKER + MINE DATA FUSION:** Combines worker status, location and mine conditions into one risk picture.
- **HUMAN-IN-THE-LOOP RESCUE:** System supports the rescue decision; the human operator remains in control.



WATCH THE MINER → SENSE THE MINE
→ SUPPORT THE RESCUE

SENSE → COMMUNICATE → ANALYZE → ALERT → RECONNAISSANCE



1



SMART MINER HELMET MODULE

MPU6050: Fall & inactivity detection
Emergency Push Button: Instant SOS
OLED + Buzzer: Local status & alerts
ESP32: Processes & transmits worker data

SURFACE CONTROL STATION & UI:

Web Dashboard: Real-time monitoring
Live Data: Worker, rover & environment status
Live Video: ESP32-CAM feed
Emergency Alerts: SOS, fall & hazard events
Decision Support: Unified view for rescue teams



RECONNAISSANCE ROVER LAYER:

ARDUINO NANO:MOTOR CONTROL + SENSOR ACQUISITION
L298N → MOTOR DRIVER
HC-SR04 → OBSTACLE DETECTION
MQ-2 → GAS/SMOKE INDICATION
DHT11 → TEMPERATURE/HUMIDITY
ESP32-CAM → VIDEO AND WIRELESS COMM



BUILT FOR A MINE. DESIGNED FOR A PROTOTYPE. READY TO SCALE.

BUILDABLE

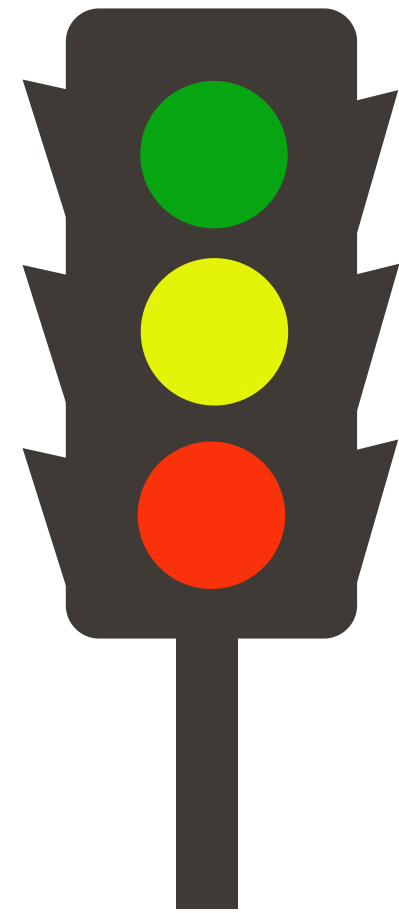
Low-cost hardware
Modular system
Rover prototype
AI dashboard

REAL-WORLD RISKS

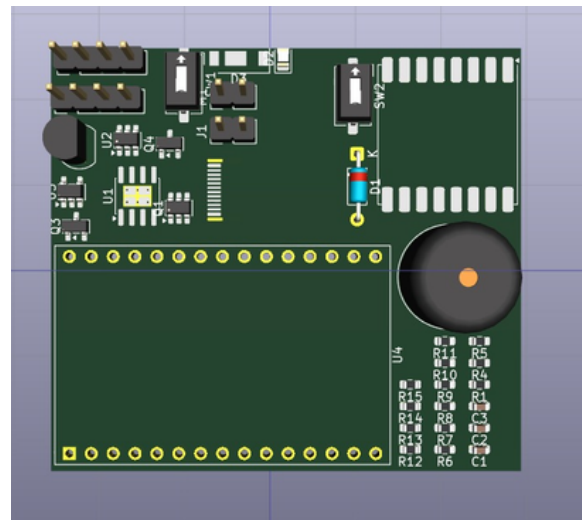
Communication
Localization
Harsh environment
False alarms

ENGINEERED MITIGATION

Relay nodes
Last-known zone
Protected enclosure
Sensor fusion



HEADGEAR CIRCUIT



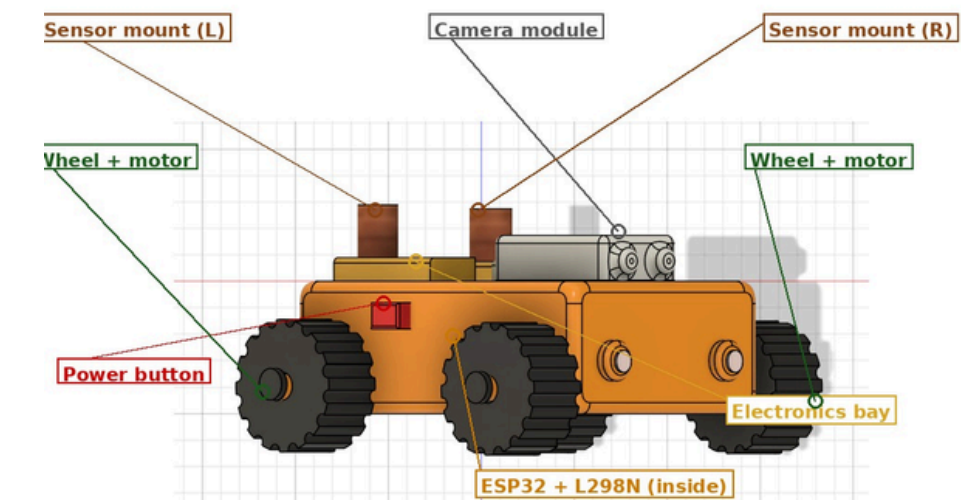
PROTOTYPE

VALIDATE

FIELD-TEST

SCALE

“We prove the critical emergency-to-rescue workflow first —
then scale with mine-grade technology.”

**ROVER**

IMPACT AND BENEFITS

FROM “WHERE IS HE?” TO “SEND HELP HERE.”

PROTECT THE WORKER : Detects falls, inactivity, abnormal conditions and SOS.

KNOW WHERE TO ACT : Identify the worker's last known underground zone.

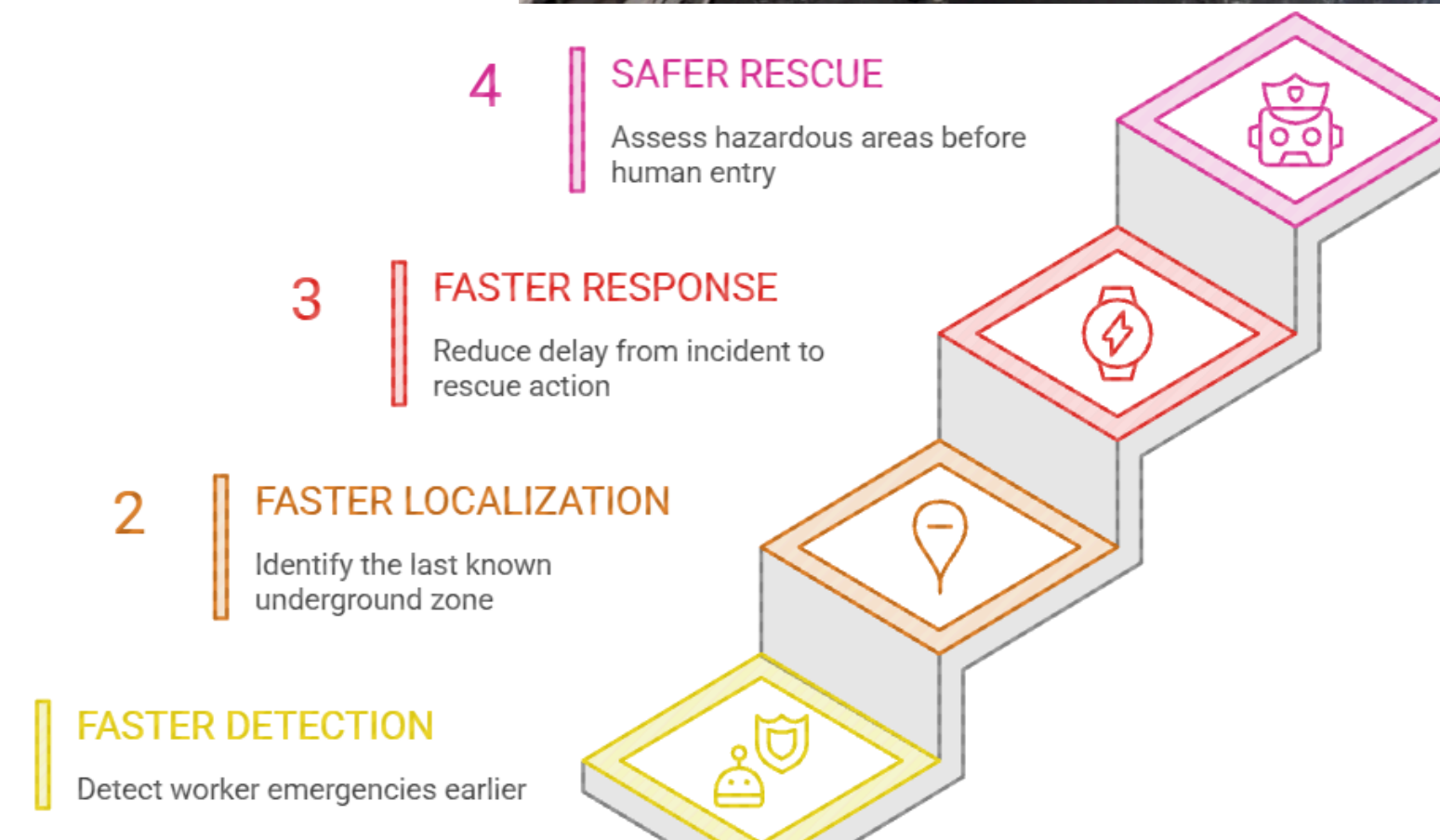
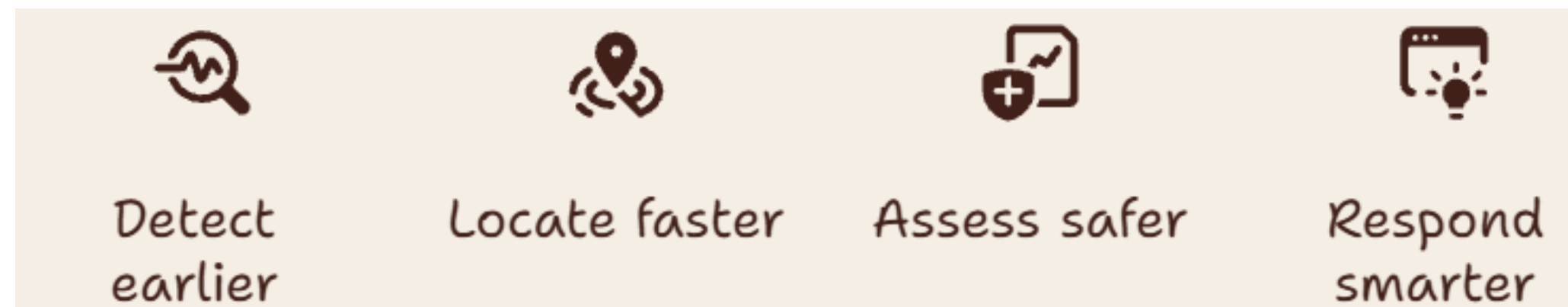
CUT RESPONSE DELAY : Automatic alerts reach the control room faster.

SAFER RESCUE : Rover-based reconnaissance checks hazards before human entry.

Worker safety • Rescue readiness • Faster decisions



WHAT WE AIM TO IMPROVE
 ↓ Detection Time | ↓ Localization Time
 ↓ Response Delay | ↑ Hazard Awareness



“When every second matters, rescuers shouldn't start by searching — they should start with a location, a condition and a plan.”

Existing Mine Rescue Robots

- Gemini-Scout — Sandia National Laboratories (USA); coal mine rescue robot combining gas sensors and thermal cameras
- CSIRO Numbat — Australia; early purpose-built mine rescue robot
- CMU Groundhog — Carnegie Mellon University; autonomous abandoned-mine mapping robot
- ANDROS Wolverine V2 — Remotec; tracked platform with improved navigation/environmental mapping
- Sub-terranean Robot (India) — Indian research explicitly built for Indian coal mine conditions.
- China University of Mining and Technology, CUMT Mine Rescue Robot / Underground Mine Robotics Research

Sensors and Hardware

- SLAMTEC, “RPLIDAR A1 Laser Range Scanner – Specifications,” SLAMTEC.
- Teledyne FLIR, “Lepton Technical Documentation,” Teledyne FLIR OEM.
- Hope Microelectronics / HOPERF, “RFM95W LoRa Transceiver Module,” HOPERF.
- Winsen Electronics, “MQ-4 Semiconductor Combustible Gas Sensor,”
- Espressif Systems, “ESP32-C3-MINI-1 Datasheet,” Espressif Systems.
- TDK InvenSense, “MPU-6050 Product Specification,” TDK InvenSense.

Power Electronics and Circuit Components

- Diodes Incorporated, “DMG9926UDM – N-Channel MOSFET,” Diodes Incorporated. Available: DMG9926UDM Product Page
- NanJing Top Power ASIC Corp., “DW01A – Battery Protection IC Datasheet.”
- TP4056, “Lithium Battery Charger IC – TP4056 Datasheet.”
- Diodes Incorporated, “AP2112K – 600mA CMOS LDO Regulator Datasheet,” Diodes Incorporated.
- USB Implementers Forum, “Universal Serial Bus Type-C Cable and Connector Specification.”

Robotics and AI

- Open Robotics, “ROS 2 Documentation,” Open Robotics. Available: ROS 2 Documentation
- Open Navigation LLC, “Nav2 – Navigation 2,” ROS 2 Navigation Stack. Available: Nav2 Documentation
- Steve Macenski et al., “Robot Operating System 2: Design, Architecture, and Uses,” ROS 2 technical documentation and associated publications.
- Ultralytics, “Ultralytics YOLO Documentation,” Ultralytics. Available: Ultralytics YOLO Documentation
- Streamlit, “Streamlit Documentation,” Streamlit. Available: Streamlit Documentation